Symmetric point sets and the Erdős–Szekeres problem for heptagons

Draft, 1 November 2026; certificate wording revised 6 November 2026

Abstract

Let $ES(7)$ be the least n such that every n points in general position in the plane contain 7 points in convex position; the Erdős–Szekeres conjecture predicts $ES(7) = 33$. We show, with computer-assisted proofs, that every set of at least 33 points in general position that is mapped into itself by an isometry or an affine bijection other than the identity contains a convex heptagon. The bound is sharp: there are centrally symmetric and mirror-symmetric 32-point sets without a convex heptagon. The theorem reduces to four cases: a half-turn and 34 points, a threefold rotation and 33 points, a fivefold rotation and 35 points, and a reflection; this reduction is checked in Lean 4. The mirror case follows from Baek's theorem on the case $a = 4$ of the Erdős–Tuza–Valtr conjecture through an elementary reduction, which is also checked in Lean with a formalization of Baek's theorem; we confirm it independently with SAT certificates. The fivefold case also follows from a computation reported by Dumitru. The threefold and central cases rest on SAT case splits with formally checked bookkeeping, written soundness proofs of the encodings, proof certificates accepted by the formally verified checker `cake_lpr` and, for most pieces, verdicts recorded when the proofs were checked. We also record the largest heptagon-free sets for each symmetry type, with their antecedents. None of this settles $ES(7)$ or improves its upper bound.

1 Introduction

Erdős and Szekeres [6] proved that $ES(k)$ is finite and constructed 2^{k-2} points in general position with no k in convex position [7]; they conjectured $ES(k) = 2^{k-2} + 1$. The conjecture holds for $k \le 6$, the case $k = 6$ by a computer search of Szekeres and Peters [14], and $ES(k) \le 2^{k+o(k)}$ by Suk [15]. The first open case is $k = 7$.

Symmetric configurations are a natural restricted search space. Subercaseaux, Mackey, Qian and Heule [16] built rotational symmetry into SAT encodings, found symmetric extremal sets for $k = 6$, reported that no real fourfold symmetric 32-point set avoids a convex heptagon, asked for a realized twofold symmetric 32-point example, and suspected that symmetric constructions exist for all $k \ge 7$. Here we show, with computer-assisted proofs, that there is no symmetric counterexample to $ES(7) = 33$ (Theorem 2); its mirror case is implied by a theorem of Baek [1], and its fivefold case also by a computation reported by Dumitru [5]. Heptagon-free 32-point sets exist with a half-turn and with a reflection, while every set of 32 or more points with a rotation of order at least 3 contains a convex heptagon (Proposition 18). Other recent computational work on $ES(7)$ [5, 10, 18, 13, 8] does not treat symmetric sets; Koshelev and Koshka [9] treat empty hexagons under axial symmetry, a different problem.

Throughout, *general position* means no three points on a line, and a *k-gon* is a set of k points in convex position. Computations are described with their trust base in Section 7; directory names

refer to `cloud/es7-2026-10-19/results` in the repository. Statements that no prior result was found refer to the sources of Section 9 only.

2 Which symmetries are possible

The symmetry group of a finite set fixes its centroid, so it is a cyclic group C_s of rotations or a dihedral group. For a rotation of order s about c, every orbit other than $\{c\}$ has exactly s points and is the vertex set of a regular s-gon.

Lemma 1. *Let S be in general position and invariant under a rotation ρ of order $s \geq 2$ about c. (a) If $s \geq 7$ and $S \neq \{c\}$, then S contains a 7-gon. (b) If s is even and $|S| \geq 3$, then $c \notin S$ and s divides $|S|$. (c) If s is odd, then $|S| \equiv 0$ or $1 \pmod{s}$.*

Proof. (a) An orbit is a convex s-gon. (b) $\rho^{s/2}$ is the half-turn, so $c, p, \rho^{s/2}(p)$ are collinear for every p; hence $c \notin S$ and all orbits have size s. (c) At most the centre is a fixed point. □

Theorem 2 (computer-assisted)**.** *Every set of at least 33 points in general position in the plane that is mapped into itself by an isometry or an affine bijection other than the identity contains 7 points in convex position.*

Proof. Let g be such a map and S the set. Since g permutes S and S spans the plane, g has finite order. An affine map of finite order becomes an isometry after a linear change of coordinates (average an inner product over the group it generates), and such a change preserves general position and convex position; so let g be an isometry. Being of finite order, g is a reflection or a rotation. Reflections are handled by Theorem 7. Otherwise a power of g is the half-turn or a rotation of prime order p. For the half-turn, $c \notin S$ and $|S|$ is even (Lemma 1(b)), hence at least 34, and 17 antipodal pairs form a set excluded by Theorem 14. For $p \geq 7$ use Lemma 1(a). For $p = 3$ or 5, delete the centre if it is in S; by Lemma 1(c) at least 33, respectively 35, points remain, and 11, respectively 7, whole orbits form a set excluded by Theorem 6, respectively Theorem 16. □

Remark 3 (Lean)*.* The deduction of Theorem 2 from five case statements, the cases of Theorems 14, 6, 16, 12 and 13, is checked in Lean 4 with Mathlib, for isometries and for affine bijections (`SYMLEAN`; axioms `propext`, `Classical.choice` and `Quot.sound`). The case statements are hypotheses there. A further file (`SYMLEAN/mirror`) proves the two mirror statements in Lean from the repository's formalization of Baek's theorem (Section 4), with the same axioms and no SAT input, so that only the half-turn, threefold and fivefold statements remain hypotheses; an independent check recompiled it. No step from a CNF formula to a case statement is checked in Lean.

Corollary 4. *Assume Theorems 6, 7 and 16 and the fourfold 32-point exclusion of Proposition 18, but not Theorem 14. Then a set of at least 33 points in general position without a convex heptagon either has no nontrivial symmetry or has symmetry group exactly the half-turn group, and then its centre is not a point of the set and its size is even, at least 34.*

Proof. As in the proof of Theorem 2; a rotation of order 4 has 8 whole orbits excluded by the fourfold exclusion, and order 6 contains order 3. The half-turn remains. □

The threefold and fivefold encodings below need a centre in general position with respect to the set.

Lemma 5 (centre perturbation)**.** *Let $m \geq 3$ be odd and let S be in general position and invariant under the rotation R of order m about $c \notin S$. Rotating each orbit about c by its own small angle gives an R-invariant set in general position whose triples have the same orientations as those of S; for generic angles no line through c contains two of its points.*

Proof. Rotations about c commute with R, so invariance is kept, and the orientations of triples of S are nonzero and depend continuously on the angles. Two points of one orbit are never collinear with c, since m is odd. For points of two different orbits, collinearity with c fixes the difference of the two angles modulo π, which generic angles avoid. □

3 Threefold symmetry

Theorem 6 (computer-assisted)**.** *Every set of* 33 *points in general position that is invariant under a rotation of order* 3 *contains* 7 *points in convex position. The largest threefold-symmetric set in general position without a convex heptagon has exactly* 31 *points.*

A threefold set has $3m$ or $3m+1$ points, so one with at least 33 points contains 11 whole orbits of size 3, a 33-point threefold set without its centre; 32 is not a possible size. An exact 31-point example (centre included) is `SYMMAX/c3_31points.txt`.

For the 33-point case, a projective normal form of the labelling (`C3-S/soundness.md`), with a generic centre by Lemma 5, gives a CNF with 168,399 variables and 4,154,921 clauses. Its unsatisfiability was established by splitting on the convex layers and on orientation bits of symmetric residues into 116,883 certified pieces. A reflection of the $H6$ strata (Lemma H6M) leaves 910 depth-2 roots. For each piece, CaDiCaL produced an LRAT proof that was accepted at solve time by the formally verified checker `cake_lpr` [17], and also by `lrat-check` as a secondary check (Section 7); these proofs were then deleted. Two independently written composers recompute the CNF hash of every certified piece from the base formula and find all 910 roots covered (`C3FINAL`); that the 73,567 leaves of one explicit cut cover all 910 roots is checked in Lean (`C3LEAN`, core Lean, `decide +kernel`). In addition, 2,120 randomly drawn pieces were re-solved from scratch with fresh proofs and both checkers (`C3SPOT`). That draw excluded the pieces recorded in σ-cube format; a separate run on 1 November re-solved the 204 of them that the cover cannot do without, on CNFs with the recorded hashes and with both checkers: all 204 were certified, and their proofs are archived (`C3PIECES`).

The closest earlier results concern other parameters: for 16 points without a convex hexagon the threefold formula is unsatisfiable [16, §6.1], [11], and a heptagon-free set of 32 points can only have 1-, 2- or 4-fold rotational symmetry [16, §6.2]. Dumitru [5, Table 1] reports unsatisfiable runs for two convex-layer templates compatible with threefold symmetry; a threefold 33-point counterexample may have any of 144 layer profiles, so these runs do not imply Theorem 6.

4 Mirror symmetry

Theorem 7. *Every mirror-symmetric set of at least* 33 *points in general position contains* 7 *points in convex position. The largest mirror-symmetric set in general position without a convex heptagon has exactly* 32 *points.*

The upper bound follows from a theorem of Baek by an elementary reduction, which we give first; Theorems 12 and 13 below are independent computer proofs. For a set with distinct x-coordinates, an *a-cap* (an *a-cup*) is a sequence of a of its points with increasing x whose consecutive slopes decrease (increase).

Theorem 8 (Baek [1, Thm. 1.6])**.** *For $n \geq 4$, every set of $\binom{n-1}{2}+2$ points in general position with distinct x-coordinates contains a 4-cap or n points in convex position.*

Lemma 9 (mirror lemma)**.** *Let μ be the reflection in the x-axis and let $c_1, \ldots, c_m$ ($m \geq 2$) be a cap with $y(c_1), y(c_m) \geq 0$. Then $y(c_i) > 0$ for $1 < i < m$, and $C \cup \mu C$, where $C = \{c_1, \ldots, c_m\}$, is in convex position; it has $2m$ points, minus one for each endpoint on the axis.*

Proof. An interior point of a cap lies strictly above the chord of its neighbours, which have height ≥ 0. Let f be the piecewise linear interpolation of the c_i on $I = [x(c_1), x(c_m)]$. It is concave, with a strict decrease of slope at each $x(c_i)$, $1 < i < m$, and $f \geq 0$. The region $Q = \{(x, y) : x \in I,\ -f(x) \leq y \leq f(x)\}$ is convex, as the intersection of a strip, the hypograph of f and the epigraph of $-f$. Its boundary is the polygon $c_1 \ldots c_m\, \mu c_m \ldots \mu c_1$ (a repeated point deleted at an end on the axis), which is x-monotone, hence simple. Every point of $C \cup \mu C$ is an extreme point of Q: the points over $x(c_1)$ and $x(c_m)$ are the endpoints of the vertical faces of Q there, and c_i for $1 < i < m$ is not the midpoint of two points of Q, by strict concavity of f at $x(c_i)$; symmetrically for μc_i. So $C \cup \mu C$ is in convex position. □

Proposition 10 (mirror reduction)**.** *Let $j \geq 2$, $k = 2j + 1$, and suppose that every set in general position with distinct x-coordinates, with no $(j+1)$-cap and no k points in convex position, has at most M points. Then every mirror-symmetric set in general position with no k points in convex position has at most $2M$ points.*

Proof. Take the mirror as the x-axis and let R and A be the points of S above it and on it. Then $|S| = 2|R| + |A|$ and $|A| \leq 2$. For $a \in A$ the points of $R \cup \{a\}$ have distinct x-coordinates: if $p \neq q$ had the same one, p, q and μp (or μq) would be three collinear points of S. A $(j+1)$-cap of R would give $2j+2$ points in convex position by Lemma 9. A $(j+1)$-cap of $R \cup \{a\}$ through a has a as an endpoint, by the first claim of Lemma 9, and gives $2j + 1 = k$ points in convex position. So $|R| \leq M$, and $|R| \leq M - 1$ if A is not empty; in both cases $|S| \leq 2M$. □

Proof of Theorem 7. Theorem 8 with $n = 7$ gives $M = \binom{6}{2} + 1 = 16$ in Proposition 10 for $j = 3$, so a mirror-symmetric heptagon-free set has at most 32 points. An exact 32-point example with 16 pairs and no point on the axis is `SYMMAX/mirror_32points.txt`. □

Remark 11. Theorem 7, including the cases of Theorems 12 and 13, is therefore a consequence of Baek's theorem. The case $n = 7$ of Theorem 8 had been verified by SAT for pseudolinear colorings by Balko and Valtr [3], and it has its own SAT check here at 17 points (`MIRRORETV`): of four encodings, two were regenerated with the recorded CNF hashes and accepted by `cake_lpr` on 5 November (`LRATCHECK20261105/recheck`); the other two were accepted by `lrat-check` only and are not counted as certified. Proposition 10 and the bound 32 are checked in Lean 4 for a reflection in an arbitrary line, with the repository's formalization of Baek's theorem (adapted from Baek's code, with bridge lemmas adapted from Schroeder's archive [13]) and no SAT input (`SYMLEAN/mirror`). A mirror-symmetric 32-point set also follows from the classical construction in Lovász's form [12, Thm. 2.6], with the cluster at angle $-\theta$ chosen as the mirror image of the cluster at θ; for $k = 7$ no cluster lies on the axis. Our example was found independently by search. No matching prior result was found for Proposition 10 or for its combination with Baek's theorem.

Theorem 12 (computer-assisted)**.** *A mirror-symmetric set of 33 points in general position with exactly one point on the axis contains a 7-gon.*

Theorem 13 (computer-assisted)**.** *A mirror-symmetric set of 34 points in general position with no point on the axis contains a 7-gon.*

These give a second proof of the upper bound in Theorem 7, independent of Baek's theorem. By general position at most two points lie on the axis. If none does, the size is even, hence at least 34, and 17 mirror pairs form an axis-free 34-point mirror set. If one or two do, 16 mirror pairs and one axis point form a 33-point mirror set with exactly one point on the axis. Theorem 12 has a further proof (`WRITEUP/MIRROR.md`) by a signotope encoding whose soundness, normal forms, case split and leaf cover are verified in Lean 4; its roughly 75,000 CNF facts carry recorded `cake_lpr` verdicts, and their proof files were not retained. The proofs of Theorems 12 and 13 by the encoding below are single CNF formulas with retained LRAT proofs accepted by `cake_lpr` and also by `lrat-check` (`SYMZ`).

The encoding. Put the axis on the y-axis, write the pairs as $L_i = (-a_i, b_i)$, $R_i = (a_i, b_i)$ with $a_i > 0$, use homogeneous vectors $v_p = (1, x_p, y_p)$, and add one element Z with vector $(0, 1, 0)$, the horizontal point at infinity. The chirotope $\chi = \operatorname{sign}\det$ of the set and Z satisfies: (i) $\det(v_Z, v_p, v_q) = y_p - y_q$, so the zero triples are exactly $\{Z, L_i, R_i\}$; (ii) the reflection acts by $\operatorname{diag}(1, -1, 1)$, which has determinant -1 and negates v_Z, so it reverses the orientation of triples of set points and preserves that of triples containing Z; (iii) $v_{R_i} - v_{L_i} = 2a_i v_Z$ with $a_i > 0$, so $\chi(Z, L_i, q) = \chi(R_i, L_i, q) = \chi(Z, R_i, q)$; (iv) the three-term Grassmann–Plücker relations hold, and the vectors are acyclic, since $w + \varepsilon x$ is positive on all of them for small $\varepsilon > 0$; (v) numbering the pairs by decreasing height gives $\det(v_Z, v_{L_i}, v_{L_j}) = b_i - b_j > 0$ for $i < j$. The CNF encodes (i)–(v), with variables for the classes of identified triples, and the absence of a 7-gon through the four-point criterion. Every real configuration therefore gives a satisfying assignment, so unsatisfiability excludes all real configurations. The converse fails: a satisfying assignment need not be realizable, and Z imposes no metric condition. For 34 points the CNF has 838,032 variables and 16,642,456 clauses and is refuted by CaDiCaL in 443 s; for 33 points with one axis point it has 695,928 variables and 14,080,656 clauses and is refuted in 964 s. The encoding was tested on real symmetric sets and on 3,300 random symmetric sets (satisfiable exactly when no K-gon exists, no mismatch), and it reproduces the mirror hexagon bound of Proposition 20.

5 Central symmetry

Theorem 14 (computer-assisted)**.** *Every centrally symmetric set of* 34 *points in general position contains* 7 *points in convex position. Hence every centrally symmetric set of at least* 34 *points does, and the largest centrally symmetric set in general position without a convex heptagon has exactly* 32 *points.*

By Lemma 1(b) the centre c is not a point of such a set, so the set consists of antipodal pairs, and deleting pairs gives the second statement; a 32-point example is Proposition 19.

The encoding. The centre is an element: $v_c = (v_p + v_{-p})/2$ gives $\chi(c, p, q) = \chi(-p, p, q)$; the half-turn acts by a matrix of determinant $+1$ that fixes v_c, so it preserves all orientations; and by general position the zero triples are exactly $\{c, p, -p\}$, so no perturbation of the centre is needed. The first point is a vertex of the convex hull, and the other pairs are represented by their point to the left of the line from c to the first point, sorted by angle about c. The base CNF (`SYMZ/zcnf.py rot 17 7 --m 2 --hull`) has 838,032 variables and 16,653,336 clauses. A written proof, refereed independently (sound, with minor gaps in the presentation), shows that every centrally symmetric 34-point set in general position without a convex heptagon, labelled in this normal form from any hull vertex, satisfies every clause (`C2ENC`). Consequently, if $Q, Q_1, \ldots, Q_n$ are cubes (sets of literals) and the base with the units of Q and the clauses $\neg Q_1, \ldots, \neg Q_n$ is unsatisfiable, then every labelled

real configuration satisfying Q satisfies some $Q \wedge Q_i$; dropping the literals of Q from each $\neg Q_i$ is equivalent under the units of Q. This is how the covers below are used.

The case split. The convex hull is a centrally symmetric polygon with 4 or 6 vertices: 8 vertices would contain a convex heptagon. In the normal form it is determined by one further pair (a parallelogram P_a, 16 cases) or two (a hexagon, 120 cases). The base with the negations of all 136 hull cubes is unsatisfiable, and all 120 hexagonal cases are unsatisfiable, each with a retained proof accepted by both checkers. Restarting the normal form at the other hull class of a parallelogram maps P_a to P_{17-a} (reduction A), so $P_1, \ldots, P_8$ suffice. Each P_a is split by the convex hull of the 30 inner points, a centrally symmetric quadrilateral or hexagon, into $\binom{15}{2} + \binom{15}{3} = 560$ subcases, with a certified cover relative to the hull cube of P_a. Reflecting the configuration and restarting the normal form maps class i to $(a - i) \bmod 17$ (reduction B) and pairs the subcases into 287 orbits, 14 of them fixed. Reductions A and B have written proofs, reviewed independently, and were also checked on the formula: each relabelling maps every base clause into the base together with the hull clauses of the case.

Closure. The 8 covers and the 8×287 orbits were closed with 2,492 solver runs: 2,385 unsatisfiable, 107 without an answer within the time limit, none satisfiable. Each unsatisfiable run produced an LRAT proof that was accepted by `cake_lpr` and also by `lrat-check` and then deleted; its proof hash, both verdicts and, except for seven pilot units, its CNF hash were recorded. All 2,589 recorded CNF hashes were rebuilt from the base, and that every orbit of every case has a closed member is checked in Lean (core Lean, `decide +kernel`; `C2FINAL`). Seven pilot units without a recorded hash were re-solved. A random sample of 24 units recorded on the cloud machines, drawn before any result was seen, was re-solved: all 24 rebuild the recorded CNF hash and both checkers accept the re-solved proofs; 23 proofs are byte-identical to the recorded ones, and one is a different proof. A second draw of 8 units was reproduced byte for byte. Further re-checks, started on 1 November (`C2RECHECK`): the 120 hexagonal proofs on hash-checked files with both checkers (120 of 120 accepted), the 58 pilot orbits of P_8 (58 of 58 certified, all byte-identical to the recorded proofs), and the 60 orbit representatives and 5 covers recorded only on the local machine (44 of 65 certified by 1 November 2026, at 19:43 UTC, all byte-identical; still running). A spot check of one unit from each of the 242 lanes not already reached by a re-solve had 81 units certified by the same time. Then 224 of the 2,304 orbit and cover facts (9.7%) had a re-solve on a hash-checked CNF with both checkers.

No matching prior result was found for Theorem 14. Subercaseaux et al. [16] examined only fourfold symmetry at 32 points and listed a twofold 32-point realization as open; Krapivin, Przybocki and Heule [10] found no realization among 200,000 abstract 32-point solutions, with no symmetry imposed.

6 Companion results

Lemma 15. *Let S be in general position and invariant under a rotation ρ of order $m \geq 4$ about a point $c \notin S$, and suppose S has no K points in convex position for some $K \leq 2m$. Then every convex layer of S is a single orbit, and c lies strictly inside each layer.*

Proof. The vertex set V of $\mathrm{conv}(S)$ is invariant under ρ, so it is a union of orbits of m points each, and it is in convex position; hence $|V| < K \leq 2m$ and V is one orbit, a regular m-gon with centre c. Apply the same argument to $S \setminus V$. □

A special case is stated in [16, §3.2]: a fourfold 16-point set without a convex hexagon has four layers of four points.

Theorem 16 (computer-assisted)**.** *Every set of at least* 35 *points in general position with a rotation of order* 5 *contains* 7 *points in convex position.*

Such a set contains a 35-point fivefold set without centre, which by Lemma 15 would be seven nested regular pentagons. The CNF of `SYM5/symcnf.py` for this case (an auxiliary centre, generic by Lemma 5) is refuted in 415 s, and a second normal form of the same encoder using Lemma 15 in 54 s; both LRAT proofs are retained and accepted by both checkers (`SYM5`).

Remark 17. Theorem 16 also follows from an earlier reported computation, which has priority. Dumitru [5, §7, Table 1] reports an unsatisfiable Kissat run (row 5^6, $H = 30$, baseline) for 33 points whose first six convex layers are pentagons, with the remaining three points inside them; his public generator, with zero offsets, adds to the base encoding only within-layer convexity and nesting. Deleting two points of the innermost pentagon of a 35-point fivefold set gives such a configuration. No proof files or logs for that run are public, so we treat it as a prior reported claim rather than an independently verified theorem.

Proposition 18. *Let $N(\Gamma)$ be the largest size of a set in general position with symmetry Γ and no* 7 *points in convex position. Then $N(C_2) = 32$, $N(C_3) = 31$, $N(C_4) = 28$, $N(C_5) = 31$, $N(C_6) = 30$ and $N(\mathit{mirror}) = 32$.*

Proof. The examples in `SYMMAX` were found by simulated annealing with exact arithmetic, and the example in `SYM2` is the construction of Proposition 19; all were checked by a dynamic program and by brute force over all 7-subsets. The upper bounds for C_2, C_3 and mirror symmetry are Theorems 14, 6 and 7. A sixfold set of 36 points contains a threefold 33-point set, so $N(C_6) = 30$. The fivefold bound is Theorem 16 and Lemma 1(c). By Lemma 1(b), $N(C_4) \le 28$ is the statement that no fourfold 32-point set avoids a convex heptagon. It was stated in [16, §6.2], after they found that all solutions of their model have one of six inner 12-point patterns; the arXiv version does not say how these patterns were shown to be unrealizable. They also report realizable 28-point restrictions. It is certified in `C4CUT`: the layered fourfold formula (Lemma 15) with six realizability cuts, each justified by a Grassmann–Plücker final-polynomial certificate on the inner 12 points verified in exact arithmetic, has an LRAT proof accepted by both checkers. □

An explicit centrally symmetric example. Baek and Balko [2, Definition 13, Lemma 14 and §6.1] describe a construction of Valtr that blows up an arbitrary set of $k-1$ points in general position into 2^{k-2} points without a convex k-gon, using clusters with cup and cap parameters $x_i = i-1$ and $y_i = k-i-1$. For odd k, a centrally symmetric base with clusters paired by the half-turn, which exchanges cups and caps and the two parameters, gives a centrally symmetric set. The following is an explicit case of this, rotated by a quarter-turn, with a direct proof. Sort points by x-coordinate. An *a-cup* is a sequence of a points whose consecutive slopes increase, an *a-cap* one whose slopes decrease; let $G(a,b)$ be the classical set of $\binom{a+b-4}{a-2}$ points in general position with no a-cup and no b-cap [6]. The half-turn $h(p) = -p$ exchanges cups and caps.

Proposition 19. *Let $k \ge 3$ be odd, $n = k-2$ and $x_i = 2i - n$. For $i < n/2$ place a copy of $G(i+2, k-i)$, flattened by $(x,y) \mapsto (\varepsilon x, \varepsilon^2 y)$, at (x_i, x_i^3), and for $i > n/2$ place the image under h of the copy at index $n-i$. For small ε the union S is a centrally symmetric set of 2^{k-2} points in general position without k points in convex position.*

Proof. Cluster B_i has no $(i+2)$-cup and no $(k-i)$-cap, and $|S| = \sum_i \binom{n}{i} = 2^{k-2}$. Three points of the cubic at $x = a, b, c$ are collinear iff $a + b + c = 0$, impossible for odd numbers. The slope between clusters $a < b$ is close to $\sigma(a,b) = x_a^2 + x_a x_b + x_b^2 \geq 1$, slopes inside clusters tend to 0, and $\sigma(b,c) - \sigma(a,b) = (x_c - x_a)(x_a + x_b + x_c)$; so for small ε the set is in general position and consecutive segments between clusters turn according to the sign of $x_a + x_b + x_c$.

Let $P \subseteq S$ be in convex position with leftmost point in B_l and rightmost point in B_r. If $l = r$ then $|P| \leq k-2$. Otherwise the lower chain is a cup in B_l (at most $l+1$ points), then single points in clusters $c_1 < \cdots < c_t$ strictly between l and r, and the upper chain is single points in clusters $d_1 < \cdots < d_s$ strictly between, then a cap in B_r (at most $k - r - 1$ points). The turn conditions reduce to $x_l + x_{c_1} + x_{c_2} > 0$ (with $c_2 = r$ if $t = 1$) and $x_{d_{s-1}} + x_{d_s} + x_r < 0$ (with $d_{s-1} = l$ if $s = 1$). If $t, s \geq 1$ and $c_1 \leq d_s$, comparing the two sums termwise gives a contradiction; so the clusters used by the two chains are distinct, $t + s \leq r - l - 1$, and $|P| \leq (l+1) + (k - r - 1) + t + s \leq k - 1$. □

For $k = 7$ this gives 32 points; an integer instance with coordinates of absolute value at most 1,250,000 is in `SYM2`, checked by a dynamic program and by brute force over all $\binom{32}{7}$ subsets. It answers the concrete realization task of [16]; it is a special case of the construction of [2], not a different general method.

For even k the cubic construction needs a cluster at the centre, and with a centre cluster of the same scale it fails. For $k = 6$ the fourfold 16-point sets of [16, App. C] and [11, Fig. 3.28] are centrally symmetric. For even $k \geq 8$, centrally symmetric sets of 2^{k-2} points without a convex k-gon are implied by [2, Lemma 14], applied to either of these sets with a half-turn-symmetric labelling and clusters paired by the half-turn; neither paper states this, and the proof of that lemma is omitted in [2]. Exact instances from this route with 64 points ($k = 8$) and 256 points ($k = 10$) were checked (`SYMEVEN`). The same directory gives a self-contained written proof, refereed, of a cubic construction with a much narrower middle cluster, with exact instances for $k = 6$, 8 and 10.

Hexagons. For $k = 6$ the largest symmetric sets without a convex hexagon have 16 points for twofold, fourfold and fivefold rotations (the fourfold and fivefold examples are in [16]; a fourfold set is twofold), 13 points for threefold rotations (15 and 16 points are excluded by retained LRAT proofs; 16 agrees with [16]), and 12 points for a reflection, by the following elementary bound.

Proposition 20. *Let $k = 2j$ be even, and let S be a mirror-symmetric set in general position with no k points in convex position. Then $|S| \leq 2\binom{3j-4}{2j-2} + 2$; in particular $|S| \leq 12$ for $k = 6$ and $|S| \leq 58 < 2^6$ for $k = 8$.*

Proof. With the mirror as the x-axis, the points above it have distinct x-coordinates. By Lemma 9 they have no j-cap; they have no k-cup, since a cup is in convex position. By the cups-caps theorem [6] there are at most $\binom{3j-4}{2j-2}$ of them, and at most two points lie on the mirror. □

For $k = 8$, Theorem 8 with $n = 8$ improves the bound to $2 \cdot 22 + 2 = 46$, a corollary of Baek's theorem; this bound is checked in Lean (`SYMLEAN/mirror`), and a 46-point example is in `MIRRORETV`.

7 Trust base

Lean-checked are: the deduction of Theorem 2 from the five case statements, for isometries and affine bijections (Remark 3); the bookkeeping of the threefold and central case splits, that is, that the listed leaves cover all 910 threefold roots and that every central orbit has a listed member; and the first proof of Theorem 12, conditional on its CNF facts. Whether a listed leaf is certified

is checked outside Lean, by recomputing its CNF hash from the base formula. The bound of Theorem 7 from the formalization of Baek's theorem, and with it the two mirror case statements, are Lean-checked as well (`SYMLEAN/mirror`).

Written proofs, audited or refereed independently but not formalized, establish the soundness of every encoding (threefold, fivefold and fourfold, mirror, central), Lemmas 5 and 15, and the normal forms and reductions of the case splits. Lemma 9 and Proposition 10 are written proofs that are also Lean-checked. Every case except mirror symmetry rests on one encoder; Theorem 16 uses two normal forms of one encoder.

Certificates are LRAT proofs accepted by `cake_lpr`, a checker verified in CakeML [17]. A verdict counts only if `cake_lpr` prints the line `s VERIFIED UNSAT`, since it exits with status 0 also when it rejects a proof; the drivers used here read that line, not the exit status. `cake_lpr` is trusted, the solver is not. Most proofs were also checked by `lrat-check` from drat-trim, which is not part of the trust base: on the current master (commit `2e3b2dc`) it accepts an empty-clause step whose hints reach no conflict, and hence invalid proofs of satisfiable formulas. This defect is publicly known: it was reported in drat-trim issue #31 (2022), fixed, reintroduced in 2024, and reported again with a one-line fix in issue #42 and pull request #43 (29 September 2026) [4]. The retained proofs were checked with a build carrying a one-line patch for this defect (`WRITEUP/CERTIFICATES.md`), but whatever the build, `lrat-check` verdicts are a secondary check only; where the text says that both checkers accepted a proof, the certificate is the `cake_lpr` verdict. Retained and accepted by `cake_lpr` (and by `lrat-check`) are the proofs of Theorems 12, 13 and 16, of the central hull cover and the 120 hexagonal hulls, of the fourfold 32-point exclusion, and of the re-solved central units; the 204 threefold pieces re-solved on 1 November are archived as well. For the 17-point case of Theorem 8, two of four encodings are certified by `cake_lpr` (5 November) and two were accepted by `lrat-check` only; this case is not needed, since the mirror bound is Lean-checked from Baek's theorem. A replay of the retained certificates with logs bound to the input hashes accepted four of them with both checkers (one of the two proofs of Theorem 16, the fourfold proof and two threefold hexagon proofs); by 1 November 2026, at 19:43 UTC, it had not been run on the mirror proofs, the other proof of Theorem 16 or the hull cover. The hull cover was replayed with both checkers later the same day (20:10 UTC).

Recorded verdicts: for the threefold pieces and most central parallelogram pieces, `cake_lpr` (and `lrat-check`) accepted the proof at solve time, and the proof was then deleted; only the CNF hash, the proof size, the proof hash (for every central record and for 26,436 threefold records) and the verdicts remain. The roughly 75,000 CNF facts of the first proof of Theorem 12 carry recorded `cake_lpr` verdicts only. These records were re-checked by random samples and targeted re-solves, not exhaustively.

8 What remains open

$ES(7)$ itself is open, and no unrestricted bound is improved here. The recorded verdicts of the threefold and central case splits can be re-checked only by re-solving the pieces, at a cost comparable to the original computations. No step from a CNF formula to a case statement is formalized. The encodings are abstract: the centre or point at infinity as an element imposes no metric condition, and a satisfying assignment need not be realizable. For example, the layered fourfold 32-point formula is satisfiable, as [16] also observed for its own model, although no real set exists. Only unsatisfiable answers, with sound encodings and certified covers, are used.

9 Sources inspected

For the statements on prior work we inspected [16] (arXiv version and public code; the text of its CICM 2025 chapter was not accessible), [11] (in part), [5] with its public generator, [1] with its Lean formalization, [2] (in part), [3], [12], [7], [10], [18] (through a cached copy), [13] (text search), [8] (Problem 6.58), [9], further arXiv papers on Erdős–Szekeres problems by Heule, Scheucher, Subercaseaux and coauthors (text search), and index searches (OpenAlex, Semantic Scholar, the arXiv API, zbMATH Open, OEIS, erdosproblems.com, MathOverflow). Not inspected: the journal versions of [2] and of Balko–Valtr, Erdős–Tuza–Valtr (1996), and Dumitru's announced second version. Priority statements are provisional.

References

[1] J. Baek, On the Erdős–Tuza–Valtr conjecture, *European J. Combin.* 124 (2025), 104085; arXiv:2206.04260. Theorem numbers refer to the arXiv version.

[2] J. Baek and M. Balko, The Erdős–Szekeres conjecture revisited, in *41st International Symposium on Computational Geometry (SoCG 2025)*, LIPIcs 332, paper 13, 2025.

[3] M. Balko and P. Valtr, A SAT attack on the Erdős–Szekeres conjecture, *Electron. Notes Discrete Math.* 49 (2015), 425–431.

[4] drat-trim issue tracker, `https://github.com/marijnheule/drat-trim`: issue #31 (22 July 2022), issue #42 and pull request #43 (29 September 2026).

[5] B. Dumitru, Notes on the 33-point Erdős–Szekeres problem, arXiv:2512.24061 (2025).

[6] P. Erdős and G. Szekeres, A combinatorial problem in geometry, *Compositio Math.* 2 (1935), 463–470.

[7] P. Erdős and G. Szekeres, On some extremum problems in elementary geometry, *Ann. Univ. Sci. Budapest. Eötvös Sect. Math.* 3–4 (1960/61), 53–62.

[8] B. Georgiev, J. Gómez-Serrano, T. Tao and A. Z. Wagner, Mathematical exploration and discovery at scale, arXiv:2511.02864 (2025).

[9] V. Koshelev and A. Koshka, Combinatorial geometry of Erdős–Szekeres type problems: SAT/ASP modeling and linear subreduction, arXiv:2604.20120 (2026).

[10] A. Krapivin, B. Przybocki and M. J. H. Heule, Toward satisfiability modulo realizability, arXiv:2607.02958 (2026).

[11] E. Mackey, Pinwheels and polygons: symmetric realizations of polygon-free point placements via SAT, M.S. thesis, Carnegie Mellon University, CMU-CS-25-108 (2025).

[12] W. Morris and V. Soltan, The Erdős–Szekeres problem on points in convex position – a survey, *Bull. Amer. Math. Soc.* 37 (2000), 437–458.

[13] M. Schroeder, Erdős–Szekeres: upper bounds for convex heptagons and octagons, Zenodo, doi:10.5281/zenodo.22842891 (2026), unreviewed.

[14] G. Szekeres and L. Peters, Computer solution to the 17-point Erdős–Szekeres problem, *ANZIAM J.* 48 (2006), 151–164.

[15] A. Suk, On the Erdős–Szekeres convex polygon problem, *J. Amer. Math. Soc.* 30 (2017), 1047–1053.

[16] B. Subercaseaux, E. Mackey, L. Qian and M. J. H. Heule, Automated symmetric constructions in discrete geometry, arXiv:2506.00224 (2025). The final version in the CICM 2025 proceedings was not accessible to us.

[17] Y. K. Tan, M. J. H. Heule and M. O. Myreen, cake_lpr: verified propagation redundancy checking in CakeML, *TACAS* 2021.

[18] Z. Zeng, H. Liu, K. Ratnavelu and S. H. Ong, One-sided parity compression for certified SAT attacks on the 33-point Erdős–Szekeres problem, SSRN 7334758 (2026).